FEASIBLE PICARD STEPS FOR DPP LIKELIHOOD

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ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: Mariet–Sra feasible-step Picard ascent conjecture for DPP likelihood (2015) *Certificate:* Exact rational 2×2 feasible update with lower likelihood.

Posed by Mariet and Sra (2015); counterexample found by GPT-5.6 (2026-08); formalized by S. Sra; audited by S. Sra.

A determinantal point process with positive definite kernel L on a ground set $\mathcal{Y}$ assigns a subset $Y \subseteq \mathcal{Y}$ probability proportional to $\det(L_Y)$, the principal minor on Y. Given observed subsets $Y_1, \ldots, Y_n$, learning the kernel L means maximizing the log-likelihood

$$\phi(L) = \sum_{i=1}^{n} \log\det(L_{Y_i}) - n \log\det(I + L),$$

whose stationarity $\nabla\phi(L) = 0$ can be written as $\Delta + L^{-1} = L^{-1}$ for the residual

$$\Delta = \frac{1}{n} \sum_{i=1}^{n} U_i (U_i^* L U_i)^{-1} U_i^* - (I + L)^{-1},$$

where U_i is the indicator matrix selecting the coordinates in Y_i. The stationarity suggests the Picard iteration $L_{k+1} = L_k + a L_k \Delta_k L_k$ of the quoted passage; a step size a is *feasible* when it keeps the iterate positive definite, which the source guarantees for $a \leq 1/(1-\gamma)$ with $\gamma = \max\{\lambda_{\min}(LZ), 1/\lambda_{\max}(I+L)\} \in [0,1]$. Ascent is proved for $a = 1$; the conjecture asks for every feasible $a \geq 1$.

Problem 1 (As posed: Z. Mariet and S. Sra (2015), §2 (unnumbered))**.** Above we showed that for $a = 1$ ascent is guaranteed. Empirically, $a > 1$ often works well; Prop. A.1 presents an easily computable upper bound on feasible a. We conjecture that for all feasible values $a \geq 1$, iteration (2.5) is guaranteed to increase the log-likelihood.
The iteration carrying the step size is $L_{k+1} = L_k + a L_k \Delta_k L_k$, displayed as (2.7) in the source; feasibility is the bound $a \leq 1/(1-\gamma)$ of Prop. A.1.

Theorem 2 ([**Disproof**]: feasible-step ascent conjecture)**.** *A feasible step with $a = 5$ can strictly decrease the DPP log-likelihood.*

Proof. Use ground set $\{1, 2\}$ and observations $\mathcal{T} = \{\{1\}, \{2\}, \{1, 2\}\}$. Let

$$L_0 = \begin{pmatrix} 3.37200647 & 2.62325460 \\ 2.62325460 & 6.07953037 \end{pmatrix},$$

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

with the displayed decimals interpreted as exact rationals of denominator 10^8. For this data,

$$\Delta(L) = \frac{1}{3}\left[\begin{pmatrix} L_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & L_{22}^{-1} \end{pmatrix} + L^{-1}\right] - (I+L)^{-1}.$$

At $a = 5$, exact rational arithmetic gives

$$L_1 = L_0 + 5L_0\Delta(L_0)L_0 \approx \begin{pmatrix} 3.1679125391632645 & 3.1681701017272776 \\ 3.1681701017272776 & 3.4400758121589554 \end{pmatrix}.$$

Both matrices are positive definite; numerically, $\det L_0 \approx 13.6187510458$ and $\det L_1 \approx 0.8605575075$. The normalized objective is $\phi(L) = \frac{1}{3}(\log L_{11} + \log L_{22} + \log\det L) - \log\det(I+L)$. The inequality $\phi(L_1) < \phi(L_0)$ is equivalent, after exponentiating and cubing, to the exact rational comparison

$$\frac{(L_1)_{11}(L_1)_{22}\det L_1}{(L_0)_{11}(L_0)_{22}\det L_0} < \left(\frac{\det(I+L_1)}{\det(I+L_0)}\right)^3.$$

Numerically, the left and right sides are respectively

$$0.03359118941299074 \quad \text{and} \quad 0.04354945027745948.$$

Their exact rational difference is positive. Hence the step is feasible but descending. □

Code and certificate

The case is the directory `counterexamples/dpp-feasible-step/` of the archive, at commit `da9318d`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/dpp-feasible-step
python verify.py
```

- Theorem 2 (result `dpp-feasible-step`): exact arithmetic.
- `artifacts/certificate.json`: Exact rational kernels L_0, L_1 and the positive rational likelihood gap.

AI usage

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT-5.6 (2026-08). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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References

Zelda Mariet and Suvrit Sra. Fixed-point algorithms for learning determinantal point processes. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *PMLR*, pages 2389–2397, 2015.